

Bayes theorem

* Rolling a dice $\Sigma 1, 2, 3, 4, 5, 6$

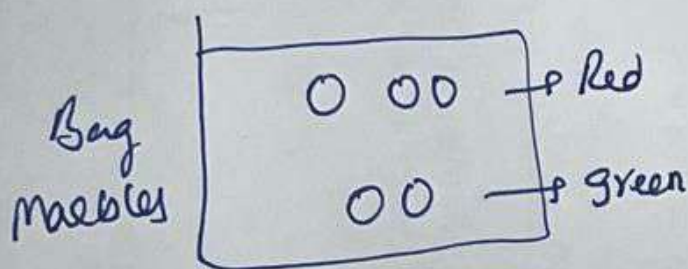
$$P(1) = \frac{1}{6}$$

$$P(2) = \frac{1}{6}$$

$$P(3) = \frac{1}{6}$$

{ Independent events }

* Dependent Event



$$P(R) = \frac{3}{5} \quad \text{First event Red}$$

$$P(G) = \frac{2}{4} \quad \text{Green event}$$

$$P(R \& G) = P(R) \cdot P(G/A)$$

Conditional probability.

$$P(A \& B) = P(A) \cdot P(B/A)$$

$$P(A \text{ and } B) = P(B \text{ and } A) \quad \text{both are same}$$

$$P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$$

$$P(B/A) = \frac{P(A/B) P(B)}{P(A)}$$

Bayes theorem

$$\begin{array}{ccccccc}
 & & & \downarrow \text{input} & & & \downarrow \text{output} \\
 x_1 & x_2 & x_3 & x_4 & x_5 & \dots & x_n & y \\
 = & = & = & = & = & = & = &
 \end{array}$$

$$P(y/x_1 x_2 x_3 \dots x_n) = \frac{P(y) * P(x_1 x_2 \dots x_n | y)}{P(x_1 x_2 \dots x_n)}$$

$$= \frac{P(y) * P(x_1 | y) * P(x_2 | y) * P(x_3 | y) \dots P(x_n | y)}{P(x_1) * P(x_2) * P(x_3) \dots P(x_n)}$$

first

x_1	x_2	x_3	x_4	y
=	=	=	=	yes
=	=	=	=	no

$$P(y = \text{yes} / x_i)$$

$$\begin{aligned}
 & P(\text{yes}) * P(x_1 | \text{yes}) * P(x_2 | \text{yes}) \\
 & * P(x_3 | \text{yes}) * P(x_4 | \text{yes}) \\
 & \hline
 & P(x_1) * P(x_2) * P(x_3) * P(x_4)
 \end{aligned}$$

$$\begin{aligned}
 P(y = \text{no} / x_i) &= P(\text{no}) * P(x_1 | \text{no}) * P(x_2 | \text{no}) \\
 & * P(x_3 | \text{no}) * P(x_4 | \text{no}) \\
 & \hline
 & P(x_1) * P(x_2) * P(x_3) * P(x_4)
 \end{aligned}$$

denominator is fixed for both of them. So constant ignore it.

NAIVE BAYES PROBLEM:

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Pb) Predicting a class label using Naive Bayesian classification.

We wish to predict the class label of a tuple using naive Bayesian classification.

Consider the following data set.

RID	Age	Income	Student	Credit-Rating	Class buys-Computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle aged	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle aged	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle aged	Medium	No	Excellent	Yes
13	Middle aged	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

In the above table, No. of distinct classes = 2 i.e. Yes and No

Classes

C1-Yes - 9

C2-No - 5

Hypothesis - H $\begin{cases} C1 \\ C2 \end{cases}$

Given Tuple is

$X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$

We need to maximize

$$P(X|C_i) P(C_i) \text{ for } i=1,2$$

$$= P(C_i/X)$$

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① → The prior probability of each class, can be computed based on the training tuples:

$$P(C_1) = P(\text{buys-Computer} = \text{yes}) = \frac{9}{14} = 0.643$$

$$P(C_2) = P(\text{buys-Computer} = \text{no}) = \frac{5}{14} = 0.357$$

② → To compute $P(x/C_i)$ for $i=1,2$,

we compute the following conditional probabilities:

$$1) P(\text{age} = \text{youth} | \text{buys-Computer} = \text{yes}) = 2/9 = 0.222$$

$$2) P(\text{age} = \text{youth} | \text{buys-Computer} = \text{no}) = 3/5 = 0.600$$

$$3) P(\text{income} = \text{medium} | \text{buys-Computer} = \text{yes}) = 4/9 = 0.444$$

$$4) P(\text{income} = \text{medium} | \text{buys-Computer} = \text{no}) = 2/5 = 0.400$$

$$5) P(\text{student} = \text{yes} | \text{buys-Computer} = \text{yes}) = 6/9 = 0.667$$

$$6) P(\text{student} = \text{yes} | \text{buys-Computer} = \text{no}) = 1/5 = 0.200$$

$$7) P(\text{credit-rating} = \text{fair} | \text{buys-Computer} = \text{yes}) = 5/9 = 0.667$$

$$8) P(\text{credit-rating} = \text{fair} | \text{buys-Computer} = \text{no}) = 2/5 = 0.400$$

Using the above $8+2=10$ probabilities, we can find

$$* P(C_i/x) = P(x/C_i) P(C_i)$$

$$* P(C_2/x) = P(x/C_2) P(C_2)$$

find
③ → $P(x/C_1)$

$$\text{We have } P(x/C_i) = \prod_{k=1}^n P(x_k/C_i) \quad n = \text{no. of attributes.}$$

$$P(x/C_1) = \prod_{k=1}^4 P(x_k/C_1)$$

$$= P(x_1/C_1) * P(x_2/C_1) * P(x_3/C_1) * P(x_4/C_1)$$

$$= P(\text{age} = \text{youth} | \text{buys-Computer} = \text{yes}) * P(\text{income} = \text{medium} | \text{buys-Computer} = \text{yes}) * P(\text{student} = \text{yes} | \text{buys-Computer} = \text{yes}) * P(\text{credit-rating} = \text{fair} | \text{buys-Computer} = \text{yes})$$

$$= 0.222 * 0.444 * 0.667 * 0.667$$

$$= 0.044$$

$$\therefore P(x/C_1) = P(x | \text{buys-Computer} = \text{yes}) = 0.044$$

② find $P(x|c_2) \equiv P(x | \text{buys-Computer} = \text{No})$

$$P(x|c_2) = \prod_{k=1}^4 P(x_k|c_2)$$

$$= P(x_1|c_2) * P(x_2|c_2) * P(x_3|c_2) * P(x_4|c_2)$$

$$= P(\text{age} = \text{youth} | \text{buys-Computer} = \text{no}) * P(\text{income} = \text{medium} | \text{buys-Computer} = \text{no}) * P(\text{student} = \text{yes} | \text{buys-Computer} = \text{no}) * P(\text{credit-rating} = \text{fair} | \text{buys-Computer} = \text{no})$$

$$= 0.600 * 0.600 * 0.200 * 0.400$$

$$= 0.019$$

$$\therefore P(x|c_2) = P(x | \text{buys-Computer} = \text{no}) = 0.019$$

$$\begin{aligned} \therefore P(x|c_1) &= 0.044 \\ P(x|c_2) &= 0.019 \end{aligned}$$

$$\therefore P(c_1/x) = P(x|c_1) \cdot P(c_1)$$

$$= P(x | \text{buys-Computer} = \text{yes}) * P(\text{buys-Computer} = \text{yes})$$

$$= 0.044 * 0.643$$

$$= 0.028 \quad \checkmark$$

$$P(c_2/x) = P(x|c_2) \cdot P(c_2)$$

$$= P(x | \text{buys-Computer} = \text{no}) * P(\text{buys-Computer} = \text{no})$$

$$= 0.019 * 0.357$$

$$= 0.007$$

$$\begin{matrix} P(c_1/x) > P(c_2/x) \\ \text{Yes} & \text{No} \end{matrix}$$

$\therefore$ The Naive Bayesian classifier predicts that buys Computer = Yes for the tuple 'x' because $P(c_1/x)$ has the maximum value out of the 2 values

$$P(c_1/x) = 0.028 > P(c_2/x) = 0.007 \therefore$$

2) The following ⁽⁶⁰⁾ table consists of training data from an employee database.

The data have been generalized. for Ex. "31...35" for age represents the age range of 31 to 35. for a given row entry, Count represents the number of data tuples having the values for department, status, age & salary in that row.

X	x_1	x_2	x_3	x_4	Y
	Department	Age	Salary	Count	Status
	Sales	31...35	46k...50k	30	Senior
	Sales	26...30	26k...30k	40	Junior
	Sales	31...35	31k...35k	40	Junior
	Systems	21...25	46k...50k	20	Junior
	Systems	31...35	66k...70k	5	Senior
	Systems	26...30	46k...50k	3	Junior
	Systems	41...45	66k...70k	3	Senior
	Marketing	26...40	46k...50k	10	Senior
	Marketing	31...35	41k...45k	4	Junior
	Secretary	46...50	36k...40k	4	Senior
	Secretary	26...30	26k...30k	6	Junior

Denominator can be neglected as the value changes proportionally.
 We need to find
 (i) $P(C_1/X)$
 i.e. $P(C_1/X) \propto P(C_1/X) \cdot P(X/C_1)$
 $P(C_1/X) = \frac{P(C_1) \cdot P(X/C_1)}{P(X)}$
 $P(C_2/X) = \frac{P(C_2) \cdot P(X/C_2)}{P(X)}$
 $C_1 = \text{Junior} = 6$
 $C_2 = \text{Senior} = 5$

Exm → Given a data tuple having the values "Systems", "26...30" & "46k...50k" & 3 for the attributes department, age, salary & Count respectively.

→ What would be the Naive Bayes classification for the tuple?

$$X = (\text{Systems}, 26...30, 46k...50k, 3)$$

Step-1 :- Find (i) $P(C_1)$
 (ii) $P(C_2)$

Here x_1, x_2, x_3 are the Categorical values.
 So we need to do the below procedure

$$X = (\text{Systems}, 26...30, 46k...50k)$$

$$(i) P(C_1) = \frac{6}{11} = 0.545$$

$$(ii) P(C_2) = \frac{5}{11} = 0.455$$

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Step-2: find i) $P(x|C_1)$ &
ii) $P(x|C_2)$

$$\textcircled{1} P(x|C_1) = \prod_{i=1}^3 P(x_i|C_1) \\ = \frac{P(x_1|C_1)}{a} * \frac{P(x_2|C_1)}{b} * \frac{P(x_3|C_1)}{c}$$

$$\textcircled{a} P(x_1|C_1) = P(\text{department} = \text{Systems} \mid \text{status} = \text{Junior}) \\ = 2/6 = 0.333$$

$$\textcircled{b} P(x_2|C_1) = P(\text{Age} = 26 \dots 30 \mid \text{status} = \text{Junior}) \\ = 3/6 = 0.5$$

$$\textcircled{c} P(x_3|C_1) = P(\text{Salary} = 46k \dots 50k \mid \text{status} = \text{Junior}) \\ = 2/6 = 0.333$$

$$= 0.333 * 0.5 * 0.333$$

$$P(x|C_1) = 0.056$$

$$\textcircled{ii} P(x|C_2) = \prod_{i=1}^3 P(x_i|C_2)$$

$$= \frac{P(x_1|C_2)}{a} * \frac{P(x_2|C_2)}{b} * \frac{P(x_3|C_2)}{c}$$

$$\textcircled{a} P(x_1|C_2) = P(\text{department} = \text{Systems} \mid \text{status} = \text{Senior}) \\ = 2/5 = 0.4$$

$$* \textcircled{b} P(x_2|C_2) = P(\text{Age} = 26 \dots 30 \mid \text{status} = \text{Senior}) \\ = 0/5 = 0$$

$$\textcircled{c} P(x_3|C_2) = P(\text{Salary} = 46k \dots 50k \mid \text{status} = \text{Senior}) \\ = 2/5 = 0.4$$

* Here $P(x_2|C_2) = 0$ - So Add One Tuple to each.

Department	Age	Salary	status
Systems	-	-	Senior
-	26...30	-	Senior
-	-	46k...50k	Senior

$$\text{Total} = 5 + 3 = 8 \text{ Senior}$$

(G2)
 $\therefore \text{New } a) P(x_1|c_2) = 3/8 = 0.375$

b) $P(x_2|c_2) = 1/8 = 0.125$

c) $P(x_3|c_2) = 3/8 = 0.375$

$\therefore P(x|c_2) = 0.375 \times 0.125 \times 0.375$

$P(x|c_2) = 0.018$

final without Count
 ① $P(c_1/x) = P(x|c_1) \times P(c_1)$
 $= 0.056 \times 0.545$

$= 0.031$

② $P(c_2/x) = P(x|c_2) \times P(c_2)$

$= 0.018 \times 0.455$

$= 0.008$

$\therefore$ Out of the 2 values $P(c_1/x)$ is having the maximum value and the corresponding class is c_1 i.e.

$\text{status} = \text{Junior}$

According to Naive Bayesian classification, the class of the tuple

x (department = Systems, Age = 26...30, Salary = 46k...50k) is

$\text{status} = \text{Junior}$

(with out Count)

$\therefore$ In the given data set also, the "status" for this tuple is "Junior" and is verified by Naive Bayesian classification. (with out Count)

Now,

Find for the Continuous Variable Count

i) $P(\text{Count} = 3 | \text{status} = \text{Junior})$

~~$P(\text{Count} = 3 | \text{status} = \text{Junior})$~~

ii) $P(\text{Count} = 3 | \text{status} = \text{Senior})$

(63)

$$P(x_k | C_i) = g(x_k, \underbrace{\mu_{Ci}}_{\text{mean}}, \underbrace{\sigma_{Ci}}_{\text{SD}}) \text{ where}$$

μ_{Ci} and σ_{Ci} are the mean and SD respectively of the values of attribute ' A_k ' for training tuples of class C_i

(i)

status	Count
Senior	30
Junior	40
Junior	40
Junior	20
Senior	5
Junior	3
Senior	3
Senior	10
Junior	4
Senior	4
Junior	6

Normal density

$$f_k(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

Junior Count
(10, 40, 20, 3, 4, 6)
↑
Mean = 18.833
SD = 15.984

$$(i) P(x_k | C_i) = P(\text{Count} = 3 | \text{class} = \text{Junior})$$

$$P(x_k | C_i) = g(x_k, \mu_{Ci}, \sigma_{Ci})$$

$$= g(3, \frac{18.833}{\mu}, \frac{15.984}{\sigma})$$

$$= \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sqrt{2\pi} (15.984)} e^{-\frac{(3 - 18.833)^2}{2(15.984)^2}}$$

$$= 0.061$$

final
with count

$$P(C_i | x) = \frac{P(x | C_i)}{a} \frac{P(C_i)}{b} = 0.0034 \times 0.545 = 0.002$$

$$(a) P(x | C_i) = P(x_1 | C_1) \times P(x_2 | C_2) \times P(x_3 | C_3) \times P(x_4 | C_4)$$

$$= 0.056 \times 0.061$$

$$= 0.0034$$

$$(b) P(C_i) = 0.545$$

(1)

$$(ii) P(x_4 | C_2) = P(\text{Count}=3 | \text{class} = \text{Senior})$$

$$P(x_4 | C_2) = g(x_k, \mu_{C_2}, \sigma_{C_2})$$

$$= g(3, 10.4, 10.092)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} * e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sqrt{2\pi * 10.092}} * e^{-\frac{(3-10.4)^2}{2(10.092)^2}}$$

$$= 0.096$$

Senior Count
20, 5, 3, 10, 4

Mean = 10.4

SD = 10.092

$$P(C_2 | x) = \frac{P(x | C_2) P(C_2)}{P(x | C_1) P(C_1) + P(x | C_2) P(C_2)} = \frac{0.002 * 0.455}{0.002 * 0.455 + 0.096 * 0.545} = 0.001$$

$$(a) P(x | C_2) = P(x_1 | C_2) * P(x_2 | C_2) * P(x_3 | C_2) * P(x_4 | C_2)$$

$$= 0.018 * 0.096$$

$$= 0.002$$

$$(b) P(C_2) = 0.455$$

∴ Of the two values $P(C_1 | x)$ and $P(C_2 | x)$, $P(C_1 | x)$ is having the maximum value and the corresponding class is C_1 i.e.

status = Junior

∴ according to Naive Bayesian classification, the class of the tuple

x (department = systems, Age = 20...30, Salary = 46k...50k, Count = 3) is

(status = Junior)

(with Count)

∴ In the given data set also, the "status" for this tuple is "Junior" and is verified by Naive Bayesian classification.